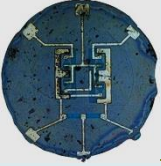


Semiconductor Physics and Devices

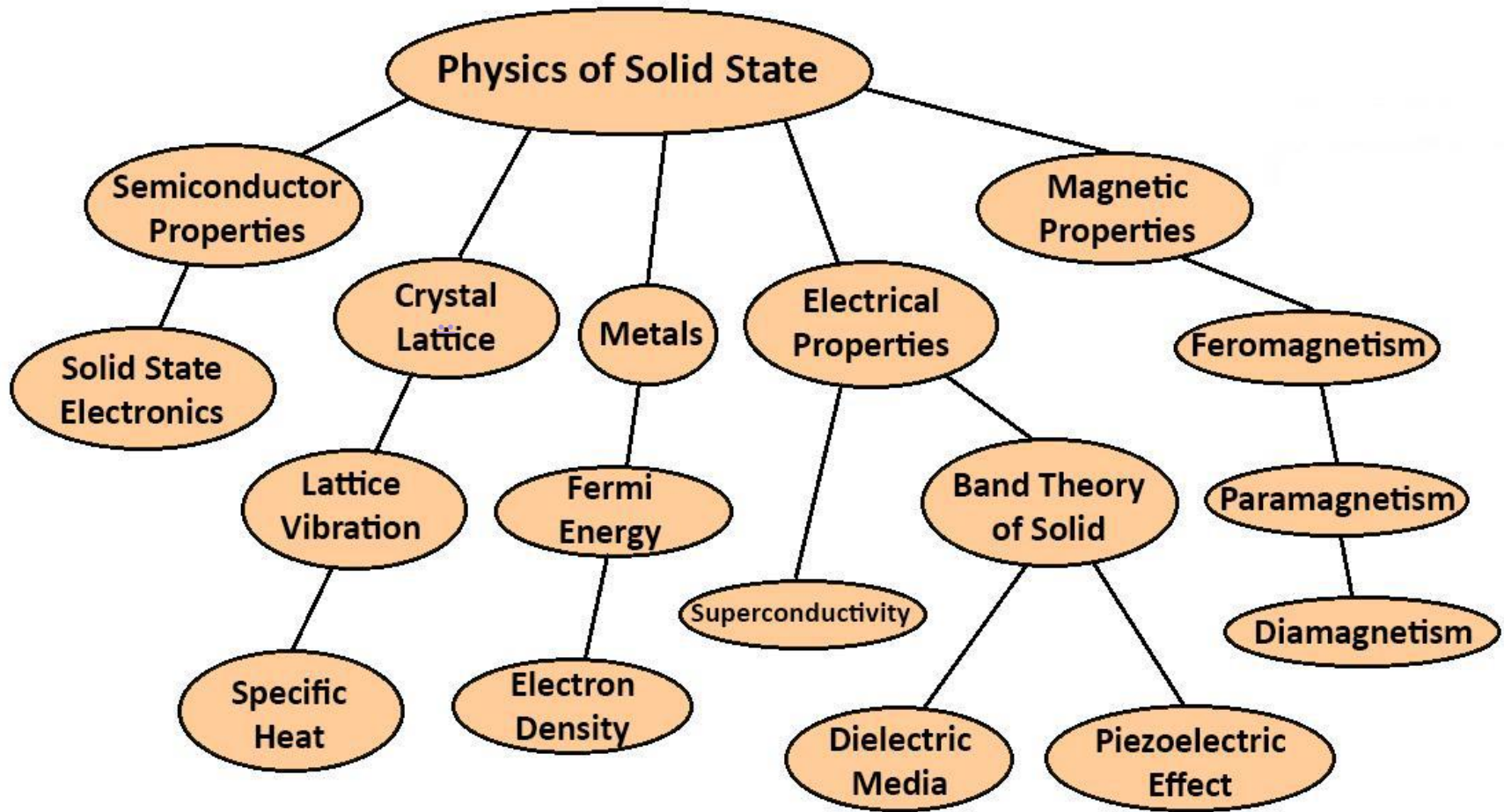
Lecture 2

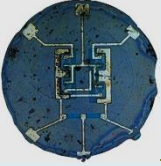
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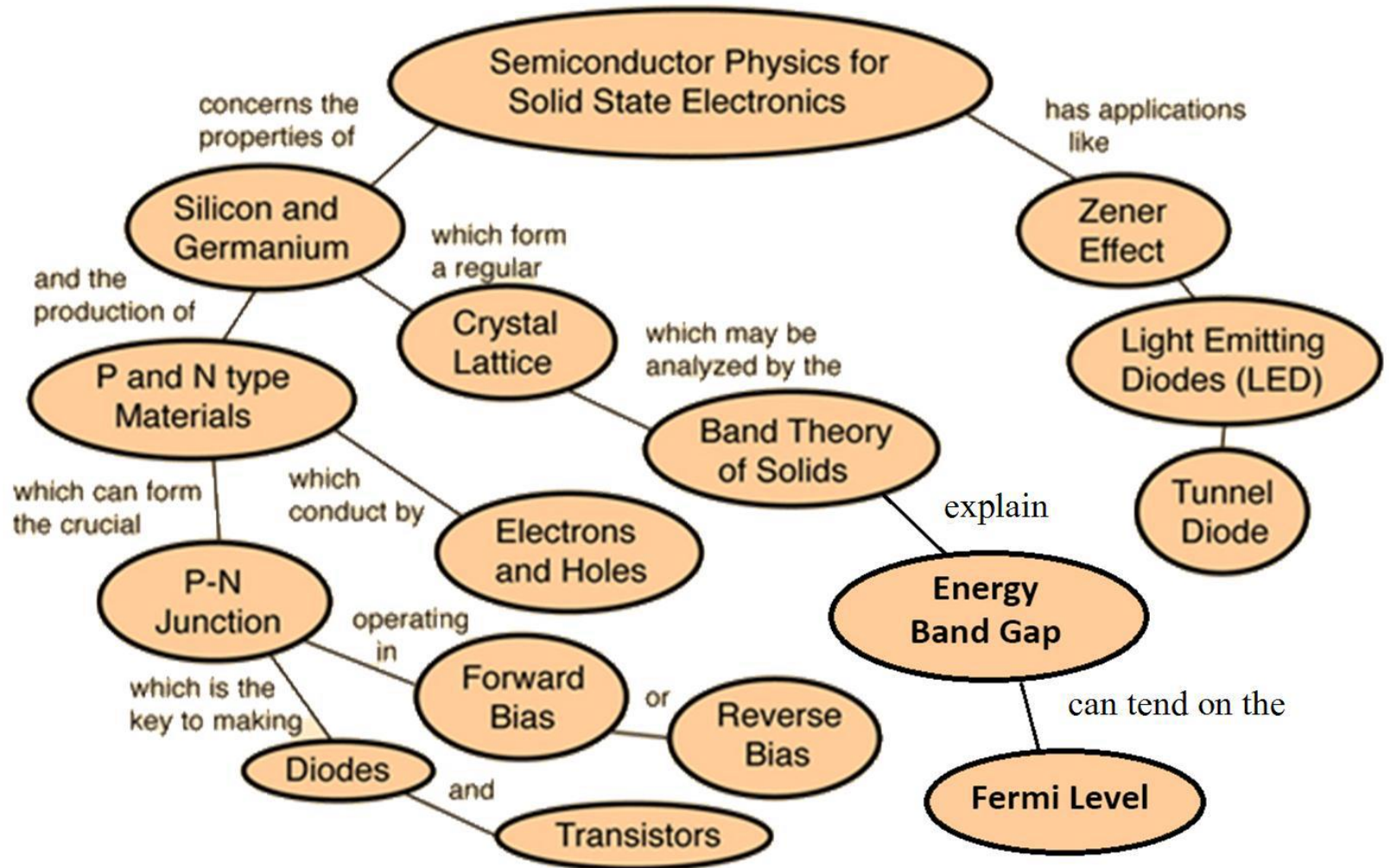


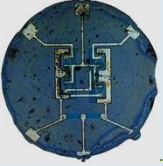
Classification Solid State Physics





Descendants of Semiconductor Physics





The nature of the atom

- ▶ In studying solid state electronics we are interested primarily in the electrical behavior of solids.
- ▶ First understand the mechanism of electron flow - in case of ac and dc

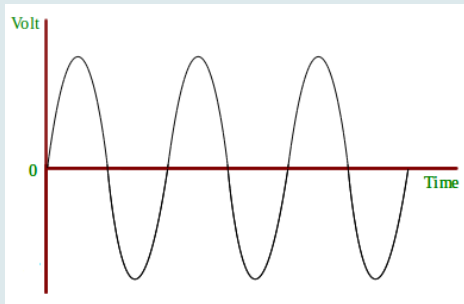


Fig: Alternating Current (AC)
Current (DC)

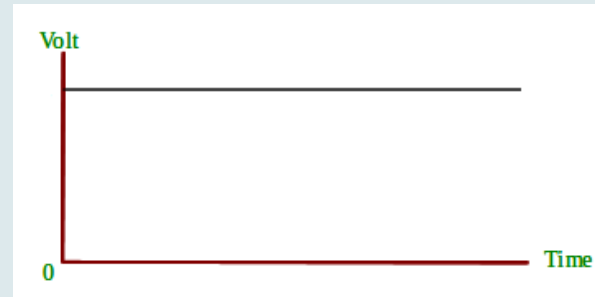
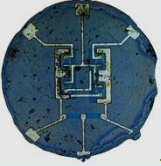


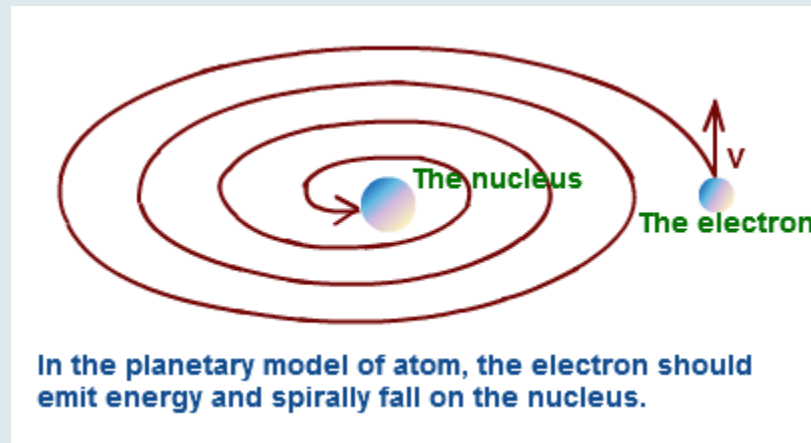
Fig: Direct

- ▶ The transport of charge through metal, semiconductor depends - not only on the properties of electron
- ▶ - but also on the arrangement of atoms in the solid.

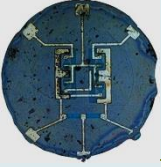


The nature of the atom

- ▶ **Rutherford in 1911 , shows**
- ▶ atom as like a planet having (+) nucleus contains nearly all mass and negatively electrons round the nucleus.



- ▶ But an accelerated charge must radiate energy in accordance with the classical laws of electromagnetism.



The nature of the atom

▶ Bohr atomic model:

- the difficulty was resolved by this model
- atom can possess only certain discrete energies/state.
- while in states electron dose not radiate energy.

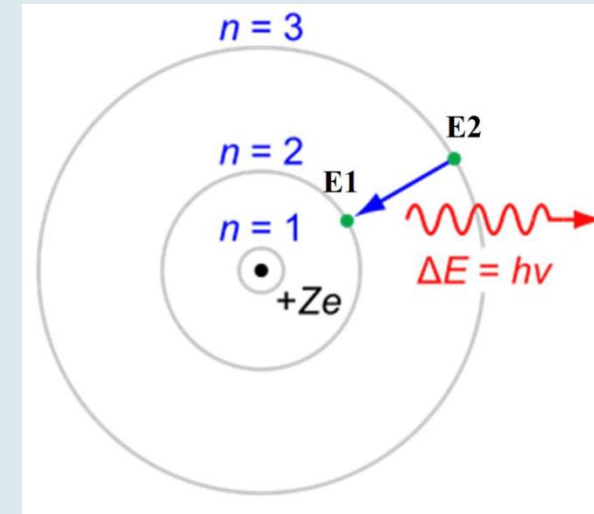
▶ $f = (w_2 - w_1)/h$

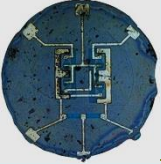
▶ $E_2 - E_1 = h\nu$

- ▶ —stationary state is determined by
- ▶ the condition

$$mvr = \frac{nh}{2\pi}$$

$$W_n = -\frac{me^4}{8h^2\epsilon_0^2} \frac{1}{n^2}$$





The nature of the atom

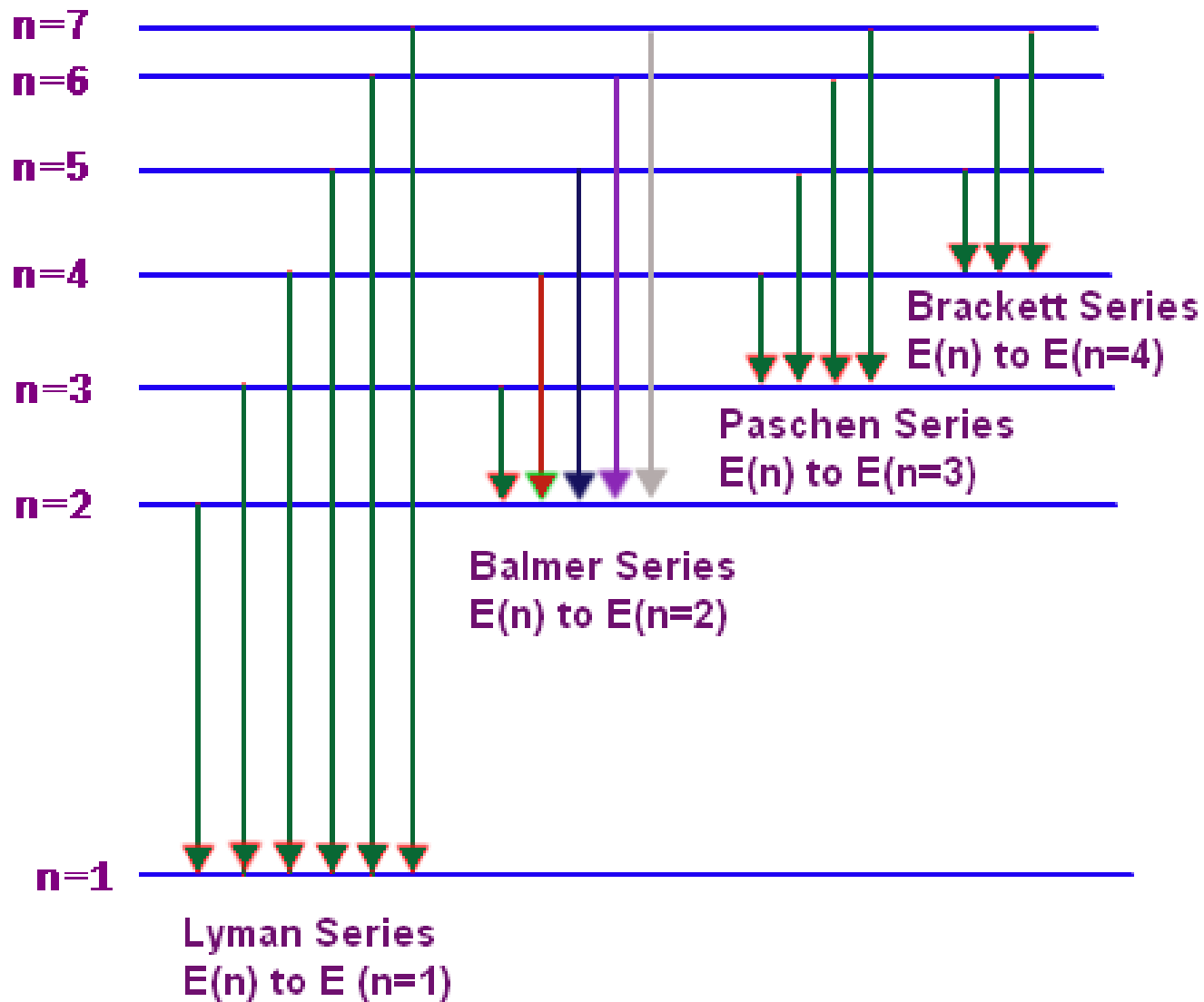
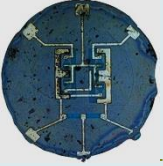
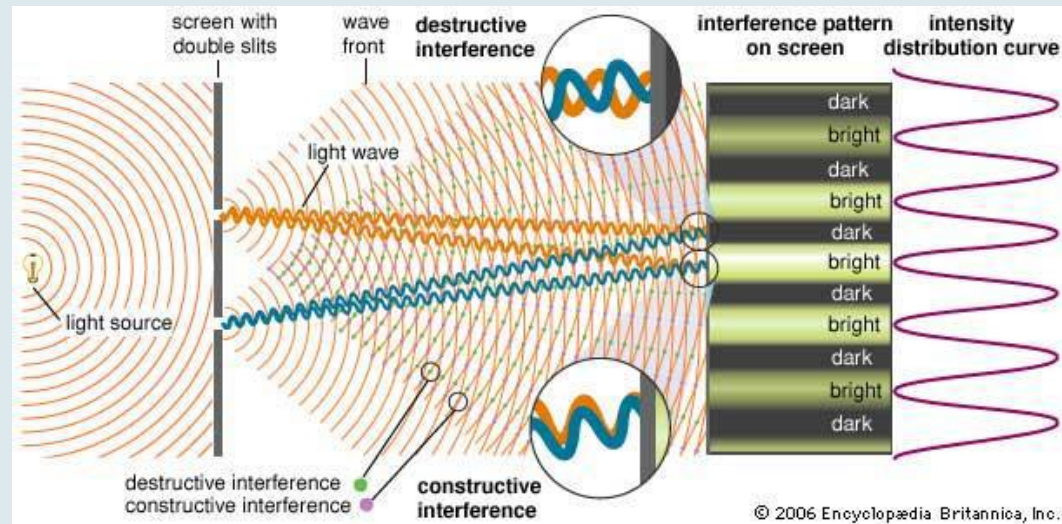


Fig: Atomic Energy Level (Spectra)



Quantum Concepts

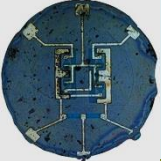
- ▶ Electrons can exhibit wave-like properties which is shown by interference and diffraction experiment like Young's double slit experiment.



Then Heisenberg uncertainty principle— it states that the more precisely the position of some particle is determined, the less precisely its momentum can be known, and vice versa

$$\Delta x \Delta p_x \geq \hbar / 2$$

$$\Delta E \Delta t \geq \hbar / 2$$



Quantum Concept

Schrodinger wave equation for one dimension is-

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\Psi = 0$$

Schrodinger wave equation for three dimension is-

$$\frac{d^2\Psi}{dx^2} + \frac{d^2\Psi}{dy^2} + \frac{d^2\Psi}{dz^2} + \frac{2m}{\hbar^2}(E - V)\Psi = 0$$

Where Ψ is the electron wave function.

The problem is that quantum concept are largely mathematical in nature and do not involve 'common sense' quality associated with classical mechanics.

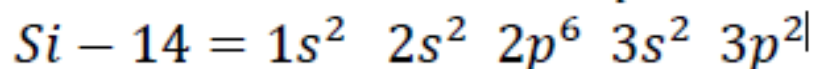
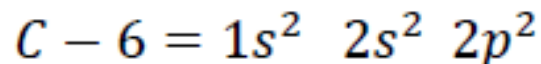
The quantum number associated with an electron—

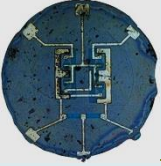
Principle quantum number, $n = 1, 2, 3 \dots \dots \dots$

Orbital angular momentum, $l = 0, 1, 2, \dots \dots \dots (n - 1)$

Orbital magnetic momentum, $m = -l, \dots \dots, -2, -1, 0, 1, 2 \dots \dots + l$

Orbital spin momentum, $s = \pm \frac{1}{2}$ for each electron.

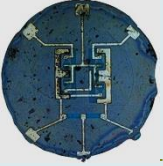




Compound State

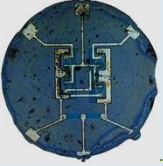
- ▶ **Solid:** The particles (ions, atoms or molecules) are packed closely together. Distance between neighboring atoms is the order of few Angstrom. The forces between particles are strong enough so that the particles cannot move freely but can only vibrate. As a result, a solid has a stable, definite shape, and a definite volume. Solids can only change their shape by force, as when broken or cut.
- ▶ **Liquid:** Intermolecular forces are week here than solid, so the molecules have enough energy to move relative to each other and the structure is mobile.
- ▶ **Gas:** In a gas, the molecules have enough kinetic energy so that the effect of intermolecular forces is small, and the typical distance between neighboring molecules is much greater (about $30\text{\AA}$) than the molecular size. A gas has no definite shape or volume, but occupies the entire container in which it is confined.





Interatomic forces and Bonding in Solids

- ▶ Many solids are aggregates of atoms. Some forces tend to hold the adjacent atoms at a particular spacing, and the process of holding them together is known as bonding.
- ▶ Electrical forces are mainly responsible in binding the atoms and molecules giving different solid structures.
- ▶ Magnetic forces have only a weak effect in cohesion.
- ▶ **The forces between atoms can be of two kinds:**
 1. Attractive forces which keep the atoms together.
 2. Repulsive forces which come into play when the solid is compressed.



Interatomic forces and Bonding in Solids

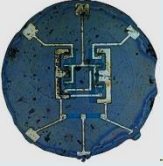
- ▶ Two atoms A and B exert attractive and repulsive forces on each other such that the bonding force F , between the atoms may be represented as-

$$F(r) = \frac{A}{r^M} - \frac{B}{r^N} \quad \text{with } N > M ;$$

Where,

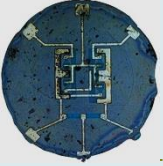
r =center to center spacing between atoms

A , B , M , N are constants characteristics of the molecule.



Solids and Its Bond

- ▶ According to the strength and directionality chemical bonds are grouped into **primary and secondary**.
- ▶ **Primary bonds: by virtue of their nature of interatomic bonds.**
- ▶ **Secondary bonds: these are intermolecular bonds.**
- ▶ —Attractive forces in primary bonds are directly associated with the valence electrons.
- ▶ —Try to keep eight electrons by taking or giving or sharing | the other shell (high energy state).



Classification of Bond

There are three strong principle types of primary bonds:

1.Ionic

2.Covalent

3.Metallic.

Van-der Waals and hydrogen bonds are typical examples of secondary bonding.

Therefore, Solids may divide into four major groups:

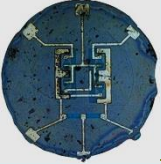
1.Ionic solid Example: *NaCl, KF*

2.Valance solid Example: *Diamond*

3.Metals Example: *Cu, Ag, Mo*

4.Van-der walls solid Example: *Ar, many organic crystals*





Ionic Solid

- ▶ Simplest type of chemical bonding.
Ionic bond is fairly strong

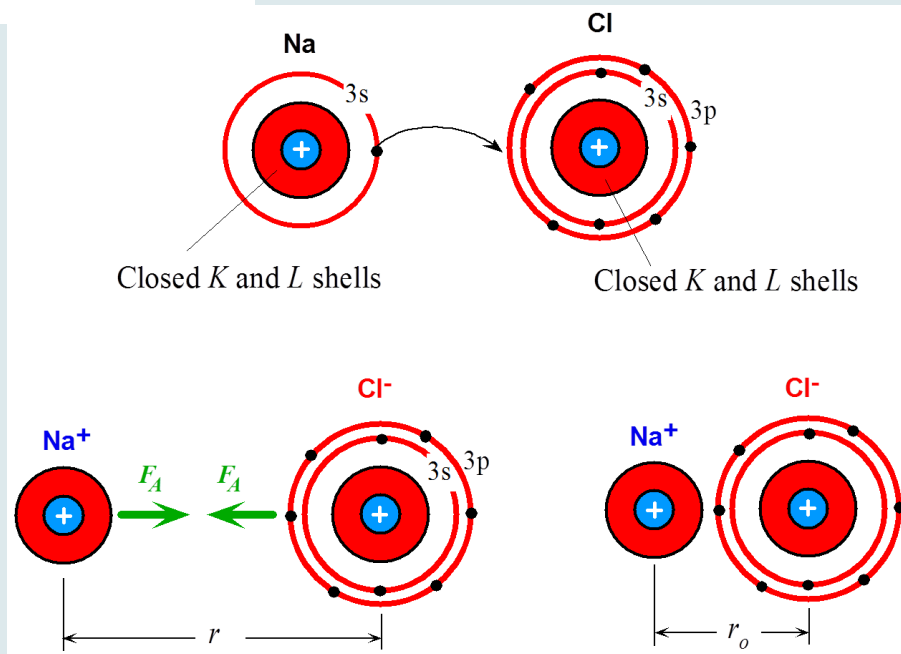
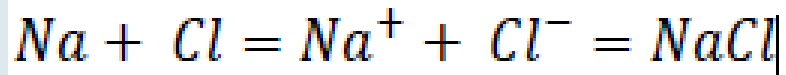
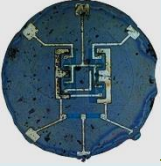


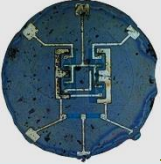
Fig: formation of ionic bond between Na and Cl

Md. Ahasan Kahir, Dept. of ETE, CUET



Properties of Ionic Solid

- ▶ Ionic compounds/solids are rigid and crystalline in nature.
- ▶ high melting points and boiling points .
- ▶ Pure and dry ionic compounds are insulators .
- ▶ Easily soluble in polar solvent and in-soluble in non-polar solvents.
- ▶ Reaction is always fast.
- ▶ High hardness and tend to cleave (break) rather than to deform.
- ▶ NaCl, MgO , CsCl, KI, LiH etc.



Covalent Solid

- ▶ By sharing of pairs of valance electrons.
- ▶ Simplest one is H_2 .
- ▶ Mostly gaseous and liquids

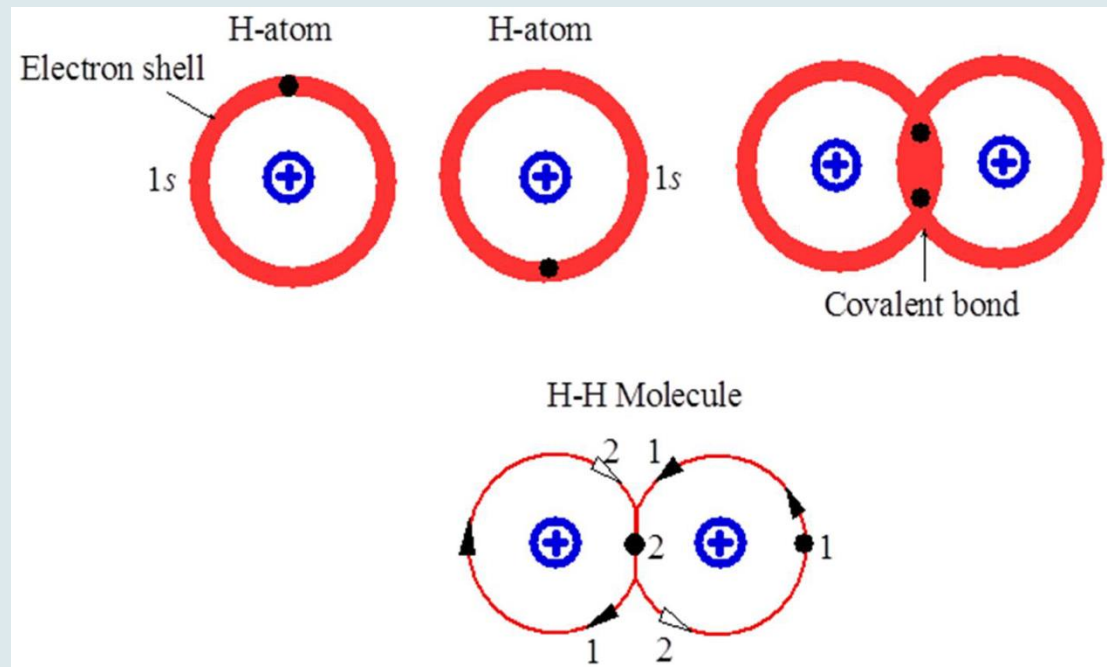
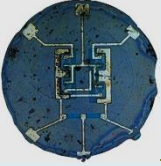
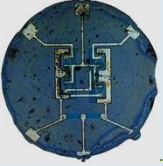


Fig: formation of covalent bond between two H atoms



Properties of Covalent Solid

- ▶ Covalent solids are very hard and brittle incapable of applicable bending.
- ▶ Melting points and boiling points are usually low as compared to those of ionic compounds.
- ▶ Poor electrical and thermal conductivity.
- ▶ Insoluble in polar solvent (H_2O) , however soluble in non-polar solvent (C_6H_6).



Metallic Bond

- ▶ The valence electrons of metal atoms are not bound to individual atoms or pair of atoms but freely move throughout the whole metal.
- ▶ It makes a electron cloud in the metal piece.
- ▶ It seems like that the positively nucleus are sink in the electron sea.
- ▶ There is a force between positively charged nucleus and free electron that is the key ingredients to form a metallic bond.

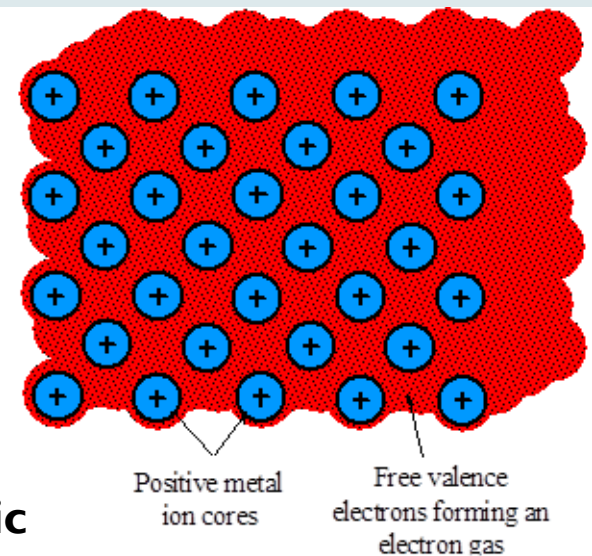
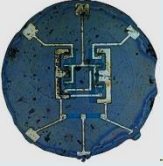
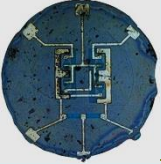


Fig: formation metallic



Properties of Metallic Bond

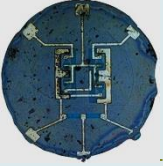
- ▶ Due to symmetrical arrangements of the positive ions in a space lattice, metals are crystalline.
- ▶ Metals have a melting point moderate to high.
- ▶ Since a large number of free electrons are available, metallic crystals have high electrical and thermal conductivity.
- ▶ Metals are opaque to light since light is absorbed by free electrons.



Van Der Waals Bond

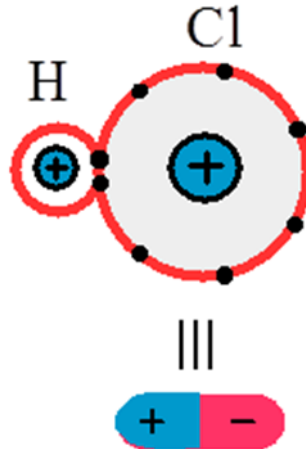
- ▶ The atoms of the rare gases such as argon have little or no tendency to give up electrons or share them with others. In the liquid and solid state the forces of attraction are the so called dispersion forces, which arise in the following way:
- ▶ The combination of the moving negative electrons and the nucleus of an atom may be considered a system of fluctuating dipoles. The interaction between these dipoles associated with neighboring atoms then gives rise to a relatively weak binding.
- ▶ In organic crystals the cohesive energy is provided by dispersion forces as well as by the interaction between permanent dipoles of neighboring molecules; the **totality of such forces** is referred to as **van der Waals forces**.



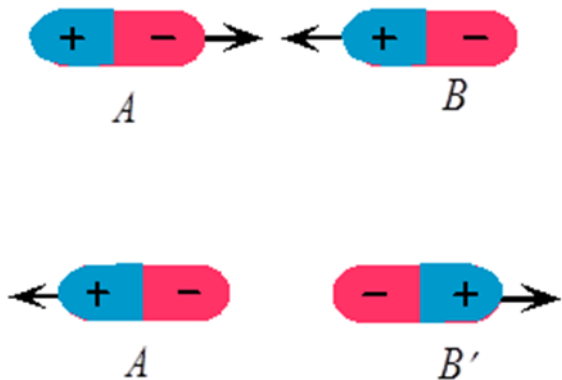


Properties of Van Der Waals Bond

- ▶ Low boiling and melting points
- ▶ Low conductivity



(a)

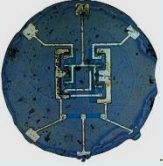


(b)



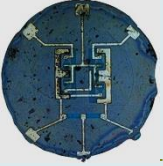
(c)

Fig: formation Van Der Waals bond.
(a) Permanently polarized molecule (dipole)
(b) Dipole can attract or repel each other
(c) Suitably oriented dipole forms van der Waals bonds



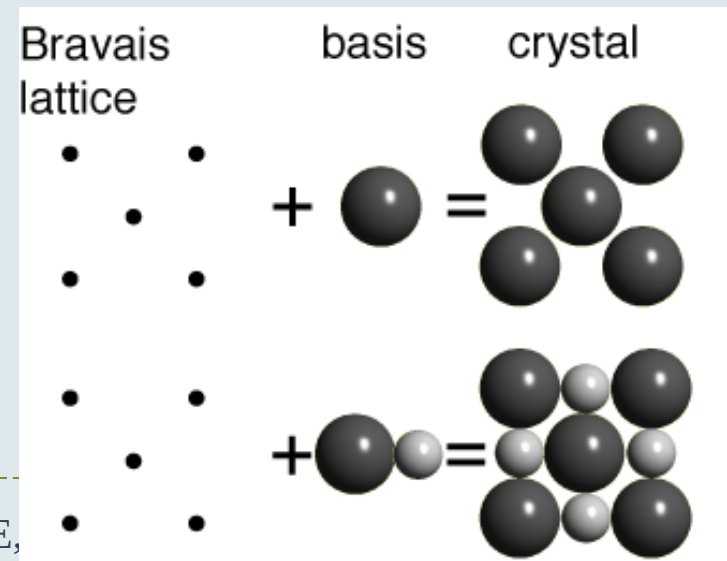
The crystalline state

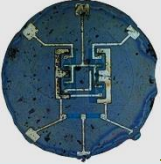
- ▶ In crystalline solids the atoms are stacked in an orderly repetitive manner based on the size of ordered regions within the materials, the solids are classified into:
 1. Amorphous: Order in amorphous solids is limited to a few molecular distances.
 2. Polycrystalline: Polycrystalline materials, the solid is made up of grains which are highly ordered crystalline regions of irregular size and orientation.
 3. Single crystal: Single crystals have long range order; extends throughout a certain piece of material.



The crystalline state

- ▶ The atomic arrangement in a crystal is called crystal structure. All crystals can be described in terms of a lattice and a basis.
- ▶ A lattice is an infinite periodic array of geometric points in space, without any atoms. (i.e. imaginary)
- ▶ When we place an identical group atoms (or molecules), called basis, at each lattice point, we obtain the actual crystal structure.
- ▶ Lattice+Basis = Crystal structure





Unit cell

- ▶ Unit cell: small groups of atoms form a repetitive pattern and the crystal properties can be described through this small unit.

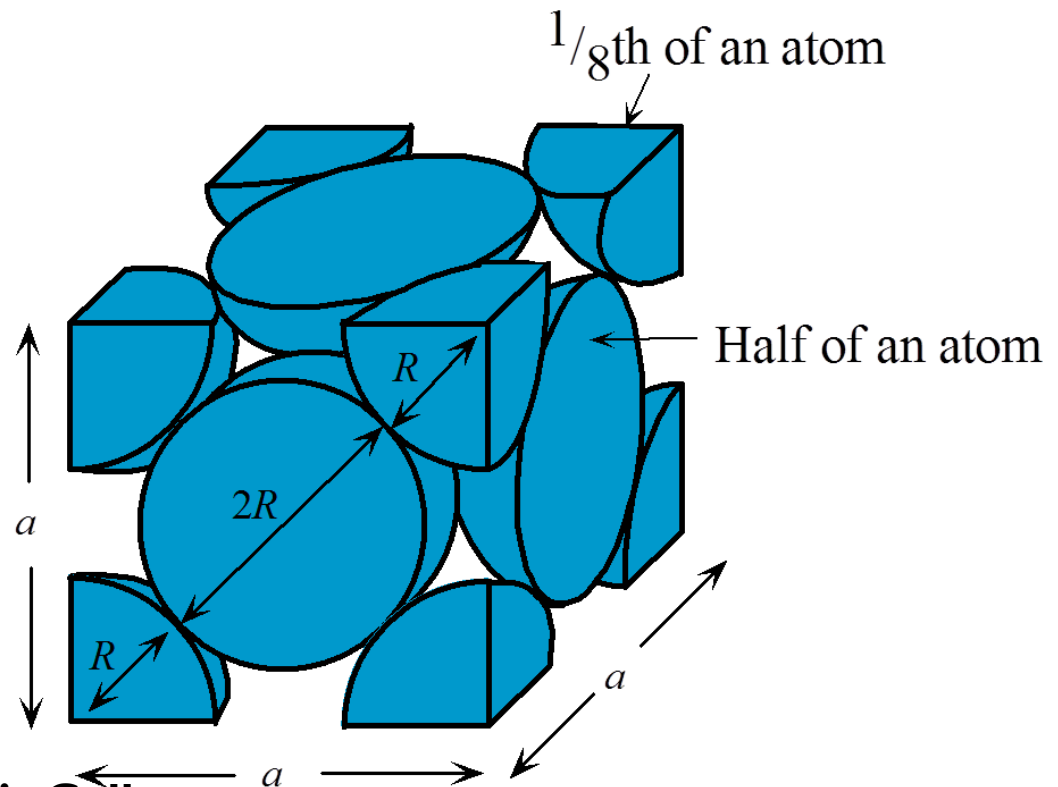
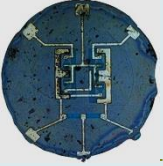


Fig: Unit Cell



Seven basic crystal systems

- Based on the length of axes and the angle between them.

Cubic:

$$a = b = c \quad \alpha = \beta = \gamma = 90^\circ$$

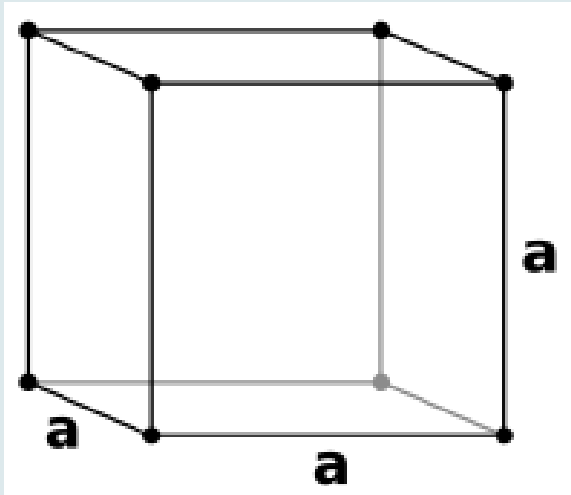
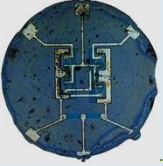


Fig: cubic structure

FeS₂



Fig: Pyrite



Seven basic crystal systems

► Tetragonal:

$$a = b \neq c \quad \alpha = \beta = \gamma = 90^\circ$$

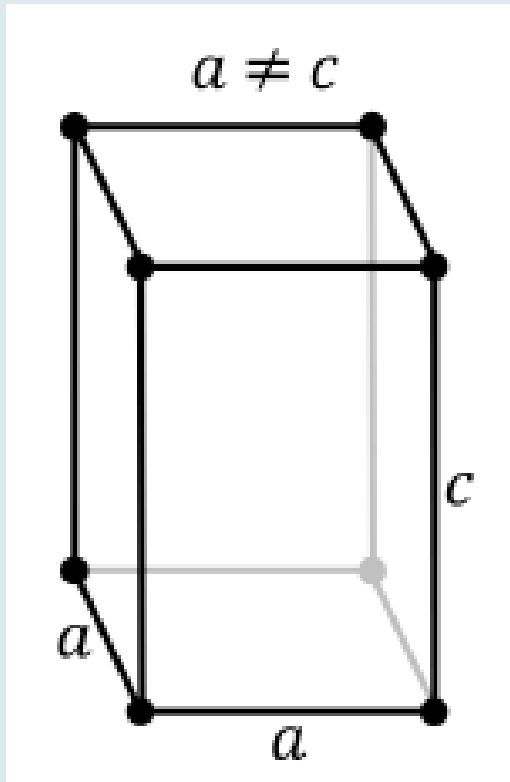


Fig: Tetragonal structure

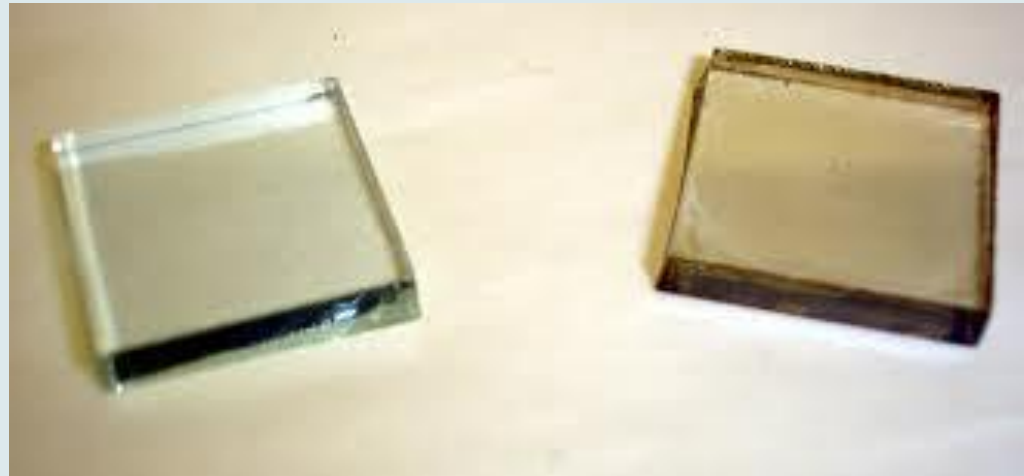
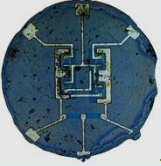


Fig: *SnO₂*



Seven basic crystal systems

- **Orthogonal:** $a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$

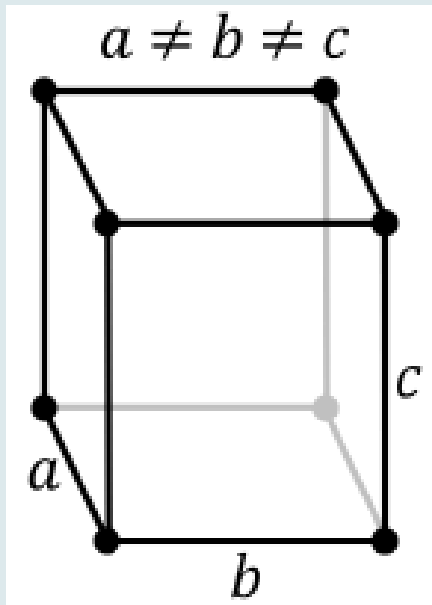
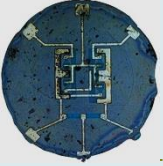


Fig: Orthogonal structure



Fig: *PbCO3*



Seven basic crystal systems

► Monoclinic:

$$a = b \neq c \quad \alpha = \beta = \gamma \neq 90^\circ$$

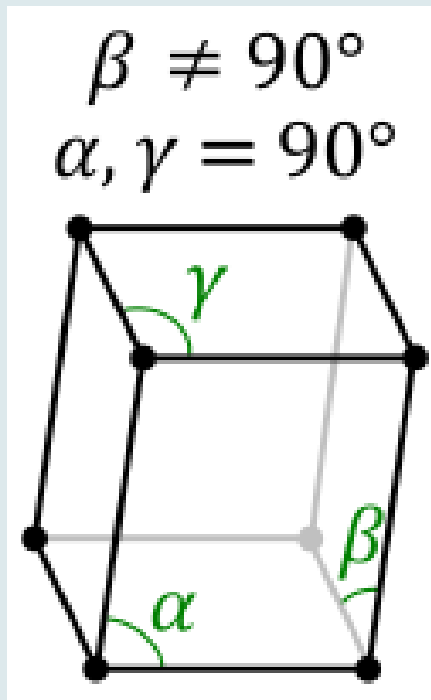
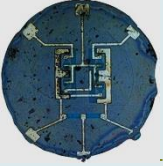


Fig: Monoclinic structure



Fig: Calcium Sulfate



Seven basic crystal systems

► **Triclinic:**

$$a = b \neq c \quad \alpha = \beta \neq \gamma \neq 90^\circ$$

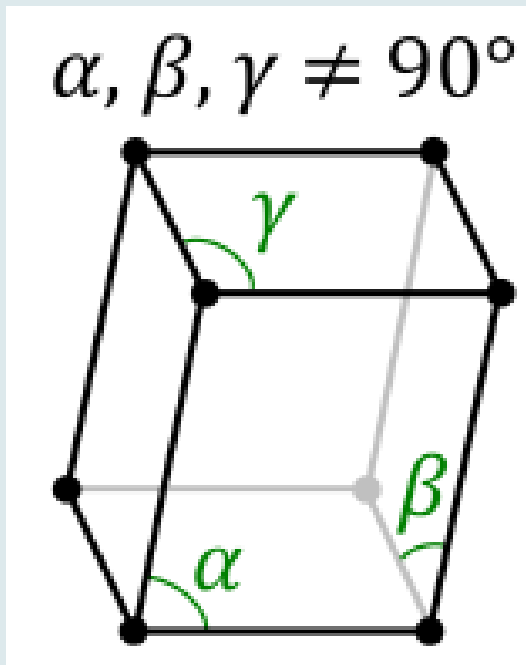
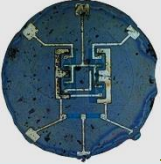


Fig: Triclinic structure
CuSO₄



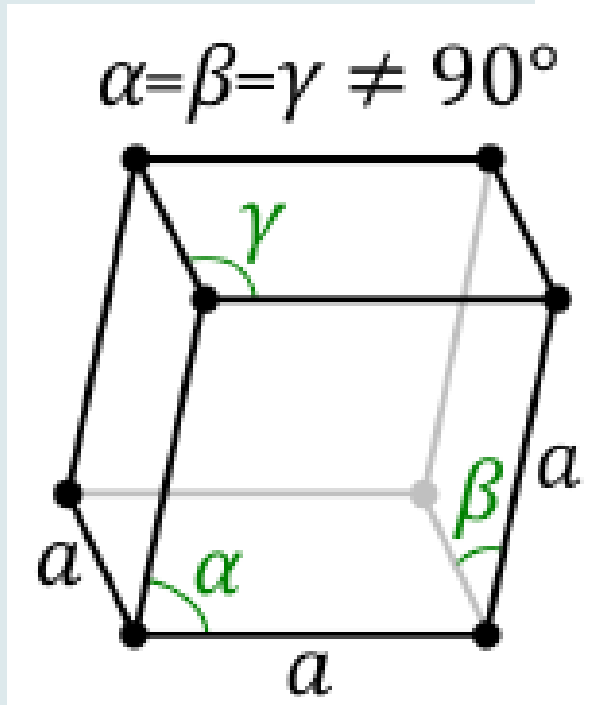
Fig: Copper Sulfate



Seven basic crystal systems

► Trigonal (Rhombohedral):

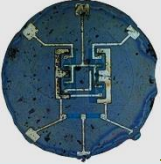
$$a = b = c \quad \alpha = \beta = \gamma \neq 90^\circ$$



**Fig: Trigonal structure
Dolomite**



Fig:



Seven basic crystal systems

- **Hexagonal:** $a = b \neq c$ $\alpha = \beta = 90^\circ, \gamma = 120^\circ$

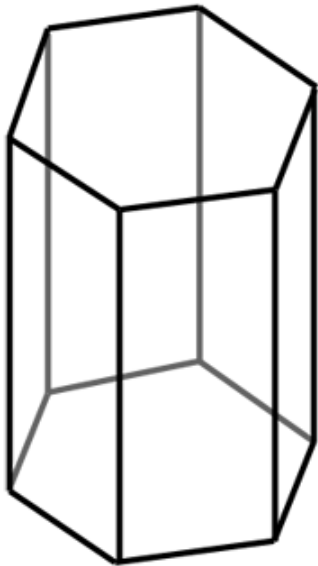
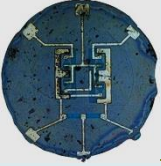


Fig: Hexagonal structure
Be₃Al₂(SiO₃)₆



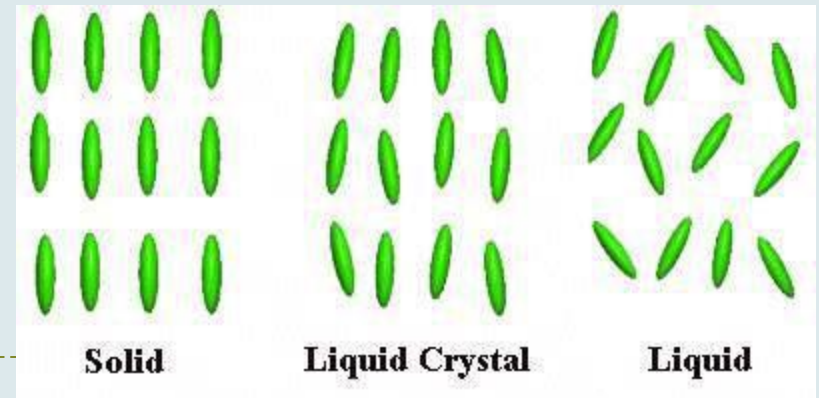
Fig: Beryl

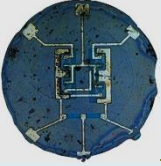


Liquid crystals

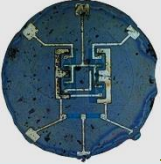
- ▶ Another class of crystals, there is two or one dimensional regularity.
- ▶ can flow and will rise in capillary tube (ammonium oleate $C_{17}H_{33}COONH_4$)
- ▶ In liquids, the atoms or molecules are in continual motion, and a crystalline structure is therefore absent.
- ▶ **Liquid crystals**
 1. Liquid crystals find wide use in liquid crystal displays
Liquid crystal in fluid form is used to detect electrically
 2. generated hot spots for failure analysis in the semiconductor industry.
 3. Used in Liquid crystal laser.

Fig: Crystal alignment





- ▶ Although one usually thinks of a solid as an arrangement of atom in which the atoms occupy fixed position relative to each other, this is not necessarily the case.
- ▶ Of course, in any crystal the atoms carry out a vibrational motion about their equilibrium position; this topic will be taken up in the next term.



Lattice: a regular geometrical arrangement of points or objects over an area or in space as the geometrical arrangement of atoms in a crystal — called also space **lattice**

The specific heat is the amount of heat per unit mass required to raise the temperature by one degree Celsius. The relationship between heat and temperature change is usually expressed in the form shown below where c is the **specific heat**.

$$Q = cm\Delta T$$

heat added specific heat mass change in temperature

The term insulator is generally used to indicate electrical obstruction while the term **dielectric** is used to indicate the energy storing capacity of the material (by means of polarization). A common example of a **dielectric** is the electrically insulating material between the metallic plates of a capacitor.

Piezoelectric Effect is the ability of certain materials to generate an electric charge in response to applied mechanical stress.

A **crystal** or **crystalline solid** is a **solid** material whose constituents (such as atoms, molecules, or ions) are arranged in a highly ordered microscopic structure, forming a **crystal** lattice that extends in all directions.